

JYOTIRMAI SINGH

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Education

Stanford University Ph.D. Physics M.S. Physics	2019 – Present 2022
University of California, Berkeley B.A. Physics <i>Highest Honors in Physics, Highest Distinction in General Scholarship, 2018 Phi Beta Kappa</i>	2015 – 2019 GPA 3.99/4.00

Research Experience

Irwin Group, Stanford University <i>Advisor: Kent Irwin</i>	09/2019 – Present Stanford, CA
- Designed and built a cryogenic superconducting resonator experiment to perform the first measurements of fundamental noise in DC SQUID amplifiers, including backaction, imprecision, and noise correlations. - Built superconducting resonators with quality factors $Q \sim 10^5 - 10^6$ for axion dark matter experiments. - Performed cryogenic noise spectroscopy to characterize microscopic dissipation channels (e.g. TLS, quasiparticles) in superconducting circuits. - Developed Squeezed Diffusion Models, a generative modeling approach inspired by quantum squeezing.	
SuperCDMS Collaboration, UC Berkeley <i>Advisor: Matt Pyle</i>	06/2018 – 05/2019 Berkeley, CA
Implemented new physics modules in the Geant4 Condensed Matter Physics package (G4CMP) to simulate phonon transport in cryogenic semiconductor crystals, advancing materials modeling capabilities for low temperature detectors	
Lawrence Berkeley National Laboratory <i>Advisor: Gabriel Orebi Gann</i>	11/2015 – 05/2019 Berkeley, CA
Developed Python analysis pipelines to incorporate uncertainties in particle position/energy reconstruction methods for neutrons linked to atmospheric neutrinos at the Sudbury Neutrino Observatory.	

Publications ([Google Scholar](#))

- Lie Group Diffusion Models for Hardware-Aware Quantum Circuit Synthesis**
J. Singh [arXiv:2606.29636 \(2026\)](#)
- Squeezed Diffusion Models**
J. Singh, S. Khanna, J. Burgess, NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences
[arXiv:2508.14871 \(2025\)](#)
- Quantum metrology of low frequency electromagnetic modes with frequency upconverters**
S. E. Kuenstner, E. C. van Assendelft, S. Chaudhuri, H. M. Cho, J. Corbin, S.W. Henderson, F. Kadribasic, D. Li, A. Phipps, N.M. Rapidis, M. Simanovskaia, J. Singh, C. Yu, K. D. Irwin, [Phys. Rev. Research 7, 013281 \(2025\)](#)
- Noise limits for dc SQUID readout of high-Q resonators below 300 MHz**
V. Ankel et al. [J. Appl. Phys. 138, 094505 \(2025\)](#)
- G4CMP: Condensed Matter Physics Simulation Using the Geant4 Toolkit**
M. H. Kelsey et al. [Nuclear Inst. and Methods in Physics Research, A 1055, 168473 \(2023\)](#)
- Measurements of DC SQUID Damping Effects on Superconducting Resonant Circuits**
E.C. van Assendelft et al. [IEEE Transactions on Applied Superconductivity \(2023\)](#)

7. **Projected Sensitivity of DMRadio-m³: A Search for the QCD Axion Below 1 μ eV**
L. Brouwer *et al.* (DMRadio Collaboration), [Phys. Rev. D 106, 103008 \(2022\)](#)
8. **Proposal for a definitive search for GUT-scale QCD axions**
L. Brouwer *et al.* (DMRadio Collaboration), [Phys. Rev. D 106, 112003 \(2022\)](#)
9. **Measurement of neutron production in atmospheric neutrino interactions at the Sudbury Neutrino Observatory**
B. Aharmim *et al.* (SNO Collaboration), [Phys. Rev. D 99, 112007 \(2019\)](#)

Skills

Programming	Python
Software	SolidWorks, COMSOL, Altium, Git
Experimental Methods	Superconducting Circuits, Cryogenic and Low-Noise Measurement (Dilution Refrigerator, Shielding/Grounding, Precision SQUID Readout), Laboratory Electronics (Oscilloscope, VNA, Lock-In Amplifiers, Waveform Generators), Machining Tools (CNC, Lathe, Bandsaw), Vacuum Equipment, Residual Gas Analyzer, Piezoelectric Positioners

Awards/Honours

Quad Fellowship (\$50,000)	2023-24
Student Presentation Award - APS Group on Instrument & Measurement Science	2021
Isidore Pomerantz Scholarship (\$1000) - Department of Physics, UC Berkeley	2018

Invited Talks & Conference Presentations

1. Squeezed Diffusion Models NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences (poster)	12/2025
2. Measurements of dc SQUID Backaction Noise and Correlations in the kHz-MHz Range APS Global Physics Summit 2025	03/2025
3. From Darkness to Light: The Search for Axion Dark Matter University of San Francisco Physics Department Colloquium	10/2024
4. LC Resonators in the DM Radio 50L Experiment APS April Meeting 2021	04/2021
5. Precision Metrology with Radiofrequency Quantum Upconverters APS March Meeting 2021	03/2021

Professional Affiliations

1. Q-NEXT National Quantum Information Science Research Center	2021 – Present
2. Kavli Institute for Particle Astrophysics and Cosmology	2021 – Present